

Sealed Experiment Record

Sealed 2026-10-07 · Signed by the K-Veritas server
Session 0e89213f · Data hash ee7e...0a94

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-07 17:11 and 17:20 UTC. Reported results: final_val_loss 1.43, final_val_ppl 4.18, arc_easy_acc 0.510522, arc_easy_acc_norm 0.455808. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	0e89213f-8afc-4f08-91f6-42ce63894838
Initialized	2026-10-07T17:11:38Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	73933f3dd35d364668fa3de8ee762ee51424a8871cbc0e3e81760acd27438f93
Disclosure	open
Source code, run 1	900a4efbaea23eb34c864a2db09bc21f0728589b36c295d0a684a43c04540d75

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	8 minutes 36 seconds	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
train_loss	1.437	3800 (76 readings)	explicit
val_loss	1.444	3200 (2 readings)	explicit
arc_easy_acc	0.510522		explicit
arc_easy_acc_norm	0.455808		explicit

Inline claim	Value	Phase	Output line
final_val_loss	1.43		3998
final_val_ppl	4.18		3999
arc_easy_acc	0.510522		4022
arc_easy_acc_norm	0.455808		4024

4 Execution evidence

Coherence

Coherent (0.47): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	5059	377.6 s	17.72 GB	85%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 199 tracked files, 4 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	17:11:38	70a76c485be601c8	199
3	run_end	17:20:14	70a76c485be601c8	0

2 snapshots with no file change omitted here; full timeline in Appendix B.

Activity: 199 files read, 0 files written, 42 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-07T17:20:15.900273+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	ee7e477d5d952aada1405b27d8779bf17729398735990b7e7175b181abe00a94
Code bundle hash	8edae1dc8e3d360285c6f32fed9e4ff8068778eb38f9cb9cf7d49979a36a1356

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
train_loss	2.992	50	3920	explicit
train_loss	2.438	100	3921	explicit
train_loss	2.466	150	3922	explicit
train_loss	2.473	200	3923	explicit
train_loss	2.416	250	3924	explicit
train_loss	2.429	300	3925	explicit
train_loss	2.129	350	3926	explicit
train_loss	1.808	400	3927	explicit
train_loss	1.874	450	3928	explicit
train_loss	1.708	500	3929	explicit
train_loss	1.604	550	3930	explicit
train_loss	1.595	600	3931	explicit
train_loss	1.434	650	3932	explicit
train_loss	1.588	700	3933	explicit
train_loss	1.445	750	3934	explicit
train_loss	1.59	800	3935	explicit
train_loss	1.393	850	3936	explicit
train_loss	1.49	900	3937	explicit
train_loss	1.417	950	3938	explicit
train_loss	1.595	1000	3939	explicit
train_loss	1.152	1050	3940	explicit
train_loss	1.254	1100	3941	explicit
train_loss	1.322	1150	3942	explicit
train_loss	1.503	1200	3943	explicit
train_loss	1.382	1250	3944	explicit
train_loss	1.631	1300	3945	explicit
train_loss	1.204	1350	3946	explicit
train_loss	1.557	1400	3947	explicit
train_loss	1.398	1450	3948	explicit
train_loss	1.414	1500	3949	explicit
train_loss	1.554	1550	3950	explicit
train_loss	1.473	1600	3951	explicit
train_loss	1.52	1650	3952	explicit
train_loss	1.414	1700	3953	explicit
train_loss	1.377	1750	3954	explicit
train_loss	1.548	1800	3955	explicit
train_loss	1.553	1850	3956	explicit
train_loss	1.347	1900	3957	explicit
train_loss	1.413	1950	3958	explicit
train_loss	1.465	2000	3959	explicit
train_loss	1.454	2050	3960	explicit
train_loss	1.541	2100	3961	explicit
train_loss	1.276	2150	3962	explicit
train_loss	1.396	2200	3963	explicit

Metric	Value	Step	Line	Source
train_loss	1.324	2250	3964	explicit
train_loss	1.37	2300	3965	explicit
train_loss	1.669	2350	3966	explicit
train_loss	1.522	2400	3967	explicit
train_loss	1.471	2450	3968	explicit
train_loss	1.486	2500	3969	explicit
train_loss	1.611	2550	3970	explicit
train_loss	1.367	2600	3971	explicit
train_loss	1.581	2650	3972	explicit
train_loss	1.38	2700	3973	explicit
train_loss	1.493	2750	3974	explicit
train_loss	1.4	2800	3975	explicit
train_loss	1.334	2850	3976	explicit
train_loss	1.413	2900	3977	explicit
train_loss	1.257	2950	3978	explicit
train_loss	1.368	3000	3979	explicit
train_loss	1.527	3050	3980	explicit
train_loss	1.514	3100	3981	explicit
train_loss	1.276	3150	3982	explicit
train_loss	1.373	3200	3983	explicit
train_loss	1.315	3250	3984	explicit
train_loss	1.429	3300	3985	explicit
train_loss	1.384	3350	3986	explicit
train_loss	1.447	3400	3987	explicit
train_loss	1.341	3450	3988	explicit
train_loss	1.351	3500	3989	explicit
train_loss	1.366	3550	3990	explicit
train_loss	1.358	3600	3991	explicit
train_loss	1.358	3650	3992	explicit
train_loss	1.504	3700	3993	explicit
train_loss	1.344	3750	3994	explicit
train_loss	1.437	3800	3995	explicit
val_loss	1.4554	1600	3996	explicit
val_loss	1.444	3200	3997	explicit
arc_easy_acc	0.510522		4021	explicit
arc_easy_acc_norm	0.455808		4023	explicit

Phase	Line	Time	CPU time	Memory	GPU
finetune	1	17:11:38	0.3 s	0.38 GB	
evaluate	4000	17:17:50	0.0 s	0.02 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	17:11:38	70a76c485be601c8	199
1	phase finetune	17:11:38	70a76c485be601c8	0
2	phase evaluate	17:17:50	70a76c485be601c8	0
3	run_end	17:20:14	70a76c485be601c8	0

Event	Path or command	Hash
read	kv_run.sh	
read	litgpt/__init__.py	
read	.devcontainer/devcontainer.json	
read	.github/ISSUE_TEMPLATE/bug-report.yaml	
read	.github/dependabot.yml	
read	.github/workflows/check-links.yml	
read	.github/workflows/cpu-tests.yml	
read	.github/workflows/mkdocs-deploy.yml	
read	.github/workflows/publish-pkg.yml	
read	.lightning/workflows/tests.yaml	
read	.pre-commit-config.yaml	
read	config_hub/finetune/falcon-7b/lora.yaml	
read	config_hub/finetune/falcon-7b/qlora.yaml	
read	config_hub/finetune/gemma-2b/full.yaml	
read	config_hub/finetune/gemma-2b/lora.yaml	
read	config_hub/finetune/gemma-2b/qlora.yaml	
read	config_hub/finetune/gemma-7b/lora.yaml	
read	config_hub/finetune/gemma-7b/qlora.yaml	
read	config_hub/finetune/gemma2-2b/lora.yaml	
read	config_hub/finetune/gemma2-2b/qlora.yaml	
read	config_hub/finetune/gemma2-9b/lora.yaml	
read	config_hub/finetune/gemma2-9b/qlora.yaml	
read	config_hub/finetune/llama-2-7b/full.yaml	
read	config_hub/finetune/llama-2-7b/lora.yaml	
read	config_hub/finetune/llama-2-7b/qlora.yaml	
read	config_hub/finetune/llama-3-8b/full.yaml	
read	config_hub/finetune/llama-3-8b/lora.yaml	
read	config_hub/finetune/llama-3-8b/qlora.yaml	
read	config_hub/finetune/llama-3.1-8b/full.yaml	
read	config_hub/finetune/llama-3.1-8b/lora.yaml	
read	config_hub/finetune/llama-3.1-8b/qlora.yaml	
read	config_hub/finetune/llama-3.2-1B/full.yaml	
read	config_hub/finetune/llama-3.2-1B/lora.yaml	
read	config_hub/finetune/llama-3.2-1B/qlora.yaml	
read	config_hub/finetune/llama-3.2-3B/full.yaml	
read	config_hub/finetune/llama-3.2-3B/lora.yaml	
read	config_hub/finetune/llama-3.2-3B/qlora.yaml	
read	config_hub/finetune/mistral-7b-v0.2/lora.yaml	
read	config_hub/finetune/mistral-7b-v0.2/qlora.yaml	
read	config_hub/finetune/mistral-7b/lora.yaml	
read	config_hub/finetune/mistral-7b/qlora.yaml	
read	config_hub/finetune/phi-2/full.yaml	
read	config_hub/finetune/phi-2/lora.yaml	
read	config_hub/finetune/phi-2/qlora.yaml	
read	config_hub/finetune/phi-3/full.yaml	
read	config_hub/finetune/phi-3/lora.yaml	
read	config_hub/finetune/phi-3/qlora.yaml	
read	config_hub/finetune/stablelm-base-alpha-3b/full.yaml	
read	config_hub/finetune/stablelm-base-alpha-3b/lora.yaml	
read	config_hub/finetune/stablelm-base-alpha-3b/qlora.yaml	

Event	Path or command	Hash
read	config_hub/finetune/tiny-llama/full.yaml	
read	config_hub/finetune/tiny-llama/lora.yaml	
read	config_hub/finetune/tiny-llama/qlora.yaml	
read	config_hub/pretrain/debug.yaml	
read	config_hub/pretrain/microllama.yaml	
read	config_hub/pretrain/tinylama.yaml	
read	config_hub/pretrain/tinystories.yaml	
read	extensions/thunder/__init__.py	
read	extensions/thunder/pretrain.py	
read	extensions/thunder/strategies/__init__.py	
read	extensions/thunder/strategies/thunder_ddp.py	
read	extensions/thunder/strategies/thunder_fsdp.py	
read	extensions/thunder/unsloth/__init__.py	
read	extensions/thunder/unsloth/executor.py	
read	extensions/thunder/unsloth/kernels/__init__.py	
read	extensions/thunder/unsloth/kernels/cross_entropy_loss.py	
read	extensions/thunder/unsloth/kernels/rope_embedding.py	
read	extensions/thunder/unsloth/kernels/swiglu.py	
read	extensions/thunder/unsloth/kernels/utils.py	
read	extensions/xla/finetune/adapter.py	
read	extensions/xla/generate/adapter.py	
read	extensions/xla/generate/base.py	
read	extensions/xla/scripts/prepare_alpaca.py	
read	extensions/xla/utils.py	
read	litgpt/__main__.py	
read	litgpt/api.py	
read	litgpt/args.py	
read	litgpt/chat/__init__.py	
read	litgpt/config.py	
read	litgpt/data/__init__.py	
read	litgpt/data/alpaca.py	
read	litgpt/data/alpaca_gpt4.py	
read	litgpt/data/base.py	
read	litgpt/data/deita.py	
read	litgpt/data/flan.py	
read	litgpt/data/longform.py	
read	litgpt/data/microllama.py	
read	litgpt/data/openwebtext.py	
read	litgpt/data/prepare_slimpajama.py	
read	litgpt/data/prepare_starcoder.py	
read	litgpt/data/text_files.py	
read	litgpt/data/tinystories.py	
read	litgpt/deploy/__init__.py	
read	litgpt/deploy/serve.py	
read	litgpt/finetune/__init__.py	
read	litgpt/finetune/adapter.py	
read	litgpt/finetune/adapter_v2.py	
read	litgpt/finetune/full.py	
read	litgpt/finetune/lora.py	
read	litgpt/finetune/lora_legacy.py	

Event	Path or command	Hash
read	litgpt/generate/adapter_v2.py	
read	litgpt/generate/full.py	
read	litgpt/generate/sequentially.py	
read	litgpt/generate/tp.py	
read	litgpt/parser_config.py	
read	litgpt/prompts.py	
read	litgpt/scripts/__init__.py	
read	litgpt/scripts/convert_hf_checkpoint.py	
read	litgpt/scripts/convert_lit_checkpoint.py	
read	litgpt/scripts/download.py	
read	litgpt/scripts/merge_lora.py	
read	litgpt/utils.py	
read	pyproject.toml	
read	tests/conftest.py	
read	tests/convert/__init__.py	
read	tests/convert/test_hf_checkpoint.py	
read	tests/convert/test_lit_checkpoint.py	
read	tests/convert/test_pretrained_checkpoint.py	
read	tests/data/__init__.py	
read	tests/data/_fixtures/alpaca.json	
read	tests/data/_fixtures/dolly.json	
read	tests/data/_fixtures/longform_train.json	
read	tests/data/_fixtures/longform_val.json	
read	tests/data/test_alpaca.py	
read	tests/data/test_base.py	
read	tests/data/test_deita.py	
read	tests/data/test_json.py	
read	tests/data/test_lit_data.py	
read	tests/data/test_longform.py	
read	tests/data/test_openwebtext.py	
read	tests/data/test_textfiles.py	
read	tests/data/test_tinyllama.py	
read	tests/data/test_tinystories.py	
read	tests/ext_thunder/__init__.py	
read	tests/ext_thunder/test_thunder_distributed.py	
read	tests/ext_thunder/test_thunder_networks.py	
read	tests/ext_thunder/test_thunder_pretrain.py	
read	tests/ext_thunder/test_unsloth_executor.py	
read	tests/generate/__init__.py	
read	tests/generate/test_adapter.py	
read	tests/generate/test_main.py	
read	tests/generate/test_sequentially.py	
read	tests/generate/test_tp.py	
read	tests/generate/utils.py	
read	tests/test_adapter.py	
read	tests/test_adapter_v2.py	
read	tests/test_api.py	
read	tests/test_args.py	
read	tests/test_batch.py	
read	tests/test_chat.py	

Event	Path or command	Hash
read	tests/test_ci.py	
read	tests/test_cli.py	
read	tests/test_config.py	
read	tests/test_config_hub.py	
read	tests/test_deepseek_moe.py	
read	tests/test_distributed.py	
read	tests/test_evaluate.py	
read	tests/test_full.py	
read	tests/test_generate_speculatively.py	
read	tests/test_lora.py	
read	tests/test_merge_lora.py	
read	tests/test_model.py	
read	tests/test_multihead_latent_attention.py	
read	tests/test_pretrain.py	
read	tests/test_prompts.py	
read	tests/test_readme.py	
read	tests/test_rope.py	
read	tests/test_serve.py	
read	tests/test_tokenizer.py	
read	tests/test_trainer_support.py	
read	tests/test_types.py	
read	tests/test_utils.py	
read	tests/test_validate.py	
read	tests/test_yarn.py	
read	tutorials/examples/ptl-trainer/litgpt_ptl_medium.py	
read	tutorials/examples/ptl-trainer/litgpt_ptl_small.py	
read	tutorials/full_finetune_example.py	
read	tutorials/mkdocs.yml	
read	litgpt/chat/base.py	
read	litgpt/model.py	
read	litgpt/constants.py	
read	litgpt/types.py	
read	litgpt/lora.py	
read	litgpt/tokenizer.py	
read	litgpt/generate/__init__.py	
read	litgpt/generate/base.py	
read	litgpt/eval/evaluate.py	
read	litgpt/adapters.py	
read	litgpt/data/alpaca_2k.py	
read	litgpt/data/json_data.py	
read	litgpt/data/lima.py	
read	litgpt/data/lit_data.py	
read	litgpt/data/tinyllama.py	
read	litgpt/adapters_v2.py	
read	litgpt/generate/adapters.py	
read	litgpt/generate/speculative_decoding.py	
read	litgpt/pretrain.py	
read	litgpt/scripts/convert_pretrained_checkpoint.py	
read	litgpt/scripts/validate.py	
process	rm -rf /var/tmp/kv-repro/015-Lightning-AI-litgpt/data/run	

Event	Path or command	Hash
process	tee /var/tmp/kv-repro/015-Lightning-AI-litgpt/finetune.out	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	python - /var/tmp/kv-repro/015-Lightning-AI-litgpt/finetune.out	
process	tail -20	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	/bin/sh -c pip list grep habana-torch-plugin	
process	grep habana-torch-plugin	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python /var/tmp/kv-repro/...	
process	pid 3888786	
process	/var/tmp/kv-repro/015-Lightning-AI-litgpt/env/bin/python -mpip list --format...	
process	pid 3893424	
process	pid 3893425	
process	python - /var/tmp/kv-repro/015-Lightning-AI-litgpt/data/run/eval/results.json	

C Run history

Invocation 1, OK, 8 minutes 36 seconds, 4024 output lines, started 2026-10-07T17:11:38

```
metric_hash 12d1b8c2ac5d3efc4ffde0a24e2ec5beef4f0f1e20e1d8ed7c5350ba6274e0e8
run_digest  92490c125e7dc31aae7e8f568adf47ecbdd1c0de60264f83f55e71b9c77aaf06
anchored    at run end
```

D Raw seal fields and hashes

Seal

```
data_hash      ee7e477d5d952aada1405b27d8779bf17729398735990b7e7175b181abe00a94
signed_message_hash 5e9e0436bddcd5dce395f885ce45586deda70ec44e2a8d6d89ed2d2d5d16a0f7
nonce          4167da9031ab8c1460d3397410841dd2
signed_at      2026-10-07T17:20:15.900273+00:00
sealed_at      2026-10-07T17:20:15.969476137Z
checkout_bundle 8edae1dc8e3d360285c6f32fed9e4ff8068778eb38f9cb9cf7d49979a36a1356
```

Signature (base64)

```
VuaDh306ymycfBK9f1qzIdHCyrjyauruQXJdID+rSr20fi7mb8ZzH2TJZ3wt+mddGMZXcmDRonoshpUJAACjIwer
8JmC5zVy0mrj/ar61mKdq2QdGHZgtuIdLDk1TytkRmjo05+pztXVWAwmDnRC1cbFzNel7ckuHvM4jzYpIercz+Sv
U0+BLA502UKGAa/n4d7WpCt2nC94uY75AI8mmLYv/aeGPHFR7AqrqtbqkRf6Xa9BM5h+umjbAWAeaAbTUSbbetF
d0Zk88s+i6qRNEXT9bZIfhav79cd1dbUgNhpr3NC5JTv+R3nC51DctlfE95uS+twfkPs0PA1MsM0PRVlHHLu44xm
el/ZyGk7AJtD60eCKpB95guo0PLHZ9kc7Jonk5IRpY0PreLJuaH3edJHEloRZZKD+S2WatfrTt5uR7q82MGkhfy7
FosgVlse2Fi8GsyRllnUxWVRjg3kx/l7nUy2ZwWdpaelLlagZ36tABrjPHvMA+xxB/dFpsWCTYwpzxKvLEEzs6ej
f3isgtzoDsuyufwysjL0cfoIKcVBL6C4L7BQ4djkwBuGU5jJwgwhByB56BUV96l6IpaGJ/f+yEE67SKQc1fZ6/KQ
KJ8nPV/000N//YC0CIK3feaptQ6HBgtvg7ruBQKfMgIX0ozZftlk7n4DEMUpfhGagAo=
```

Public key

```
-----BEGIN PUBLIC KEY-----
MIICIjANBgqhkiG9w0BAQEFAAOCAg8AMIICCGKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcProPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4LAT4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNNpXa5X
7qU95EFEisX/GfoS9ToxtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KICeadbIbHN
av0JjdTJ//CM0WZSIbdzE6qvEbNfcqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHq+D
AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9Il5xEaJn4bku4eKqeS7VSZt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPGpZ242NH8JgE6CNF7ToL9syans67u7UMgIcNgeNHnCy9bLgPU5annzP9gN2l
6HEMldkXNdKQRb1rDWyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7lE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCKDxoehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdW3gHLcCAwEAAQ==
-----END PUBLIC KEY-----
```

Run 1 outputs

```
stdout 6e06185c9db8fca188a20cf65440c2651b7b4e3aa6b159dfdd2626d255eb7719
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics 12d1b8c2ac5d3efc4ffde0a24e2ec5beef4f0f1e20e1d8ed7c5350ba6274e0e8
digest 92490c125e7dc31aae7e8f568adf47ecbdd1c0de60264f83f55e71b9c77aaf06
kv_run.sh (unchanged)
pre e686966a56678ebe9d56a05093d58eb73a0da0cace0a0e4a471de95874f30909
post e686966a56678ebe9d56a05093d58eb73a0da0cace0a0e4a471de95874f30909
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```